Test Log Template
AI_A2_2527419_TEST_LOG.pdf

Test Run Command
```bash
pytest -q
```

Expected Output
```

..................    [100%]
22 passed, 2 warnings in 7.07s
```

Test Categories Covered
1. **Data Generation** (4 tests)
- Reproducibility
- Fingerprint stability
- Schema validation
- Missing values pattern
- Duplicate detection

2. **Preprocessing** (2 tests)
- Train/test split no leakage
- Same transform train/test

3. **CLI Prediction** (2 tests)
- Valid inference
- Verification code in output

4. **API Endpoints** (7 tests)
- Health endpoint
- Model info endpoint
- Predict valid input
- Predict missing field (422)
- Predict wrong type (422)
- Predict range error (422)
- Predict invalid season (422)
- Predict invalid cooperative (422)

5. **Reproducibility** (2 tests)
- Run all reproducible
- Models load consistently

6. **Metrics Artifacts** (3 tests)
- Regression metrics structure

- Classification metrics structure
- Clustering report structure

Failure Path Tests
- All 422 validation errors exercise failure paths
- Missing field, wrong type, range error, invalid enum values

Happy Path Tests
- Valid inference through CLI and API
- Health and model-info endpoints